

Lecture 11 to Lecture 38 is *very very important*** to cover. Rest all is somewhere advance concepts, but you can watch if want to build high level of applications.**

Lecture 11 (22nd June, 2021) -

https://drive.google.com/file/d/1B_LkxXEKEE-oPmxc1JgCpy_oF7HAZ7oC/view

Topics covered:

Interface

Typescript

Lecture 12 (23rd June, 2021) -

<https://drive.google.com/file/d/1LtEiwJRI-NWngjhALFnAWzUSY5HkZKoO/view>

Topics covered: **Continued Typescript** functions and anonymous functions (11:10 am) objects with and without classes, interfaces for type checking (11:44 am) constructor, optional properties and exclamatory mark (12:25 pm) optional chaining (01:00 pm) &&, ||, and falsy values (01:10 pm)

Lecture 13 (24th June, 2021) -

<https://drive.google.com/file/d/1lzkLfsp8cxnakZdahOG7fpVAluVLgUKn/view> Topics

covered: Union type (11:10am) How code is read in JavaScript creating new types (11:30 am) Types, interfaces, classes type casting (11:44 am) Enum (11:58 am) Exception handling (12:07pm)

Lecture 14 (25th June, 2021) -

<https://drive.google.com/file/d/1ZWGAly91fVDnRawsBtLRKMpENopT4s1K/view> Topics covered: 1. Generics in OOPS 2. Fat arrow functions 3. React + Tailwind setup

Lecture 15 (28th June, 2021) -

<https://drive.google.com/file/d/1GgusQPfjleptknW0yPTkSzi7aJlwnG/view> Topics

covered: 1. Reusable React components and props 2. childrens prop 3. Aspect ratio boxes 4. How to use images in React

Lecture 16 (29th June, 2021) -

<https://drive.google.com/file/d/15rjvNZWVGcQ1QIHxm8af0c2fg1sXUe1o/view> Topics covered: 1. Object destructuring, structuring and spread operator. 2. Intro to Server side rendering vs Client side rendering and its impact on SEO and social sharing. 3. Intro to static site generator Gatsby. 4. Imports and exports in JS.

Lecture 17 (1st July, 2021) -

<https://drive.google.com/file/d/1xNGFg3eOfSN2ce2us1qxEba34CWUL43a/view> Topics

covered: 1. How to setup React project with TypeScript. 2. How to create React components in TypeScript. 3. Use of defaultProps. 4. Creation and usage of VS code snippets.

Lecture 18 (2nd July, 2021) -

<https://drive.google.com/file/d/1HXZvfin9rbhAl8jktHrxtLC5m49t42uC/view> Topics covered: 1. Difference between imperative and declarative approach. 2. A bit about React virtual dom. 3. JavaScript Tuples 4. Tuple and Array destructuring. 5. How state in React works and why it's needed.

Lecture 19 (5th July, 2021) -

<https://drive.google.com/file/d/1pqP-Zvzq02T85B72aBq4O4hojCv-Y3DR/view> Topics covered: 1. Continued React State. 2. React.memo 3. React Router

Lecture 20 (6th July, 2021) -

<https://drive.google.com/file/d/1svnhK3Jqz2aL6SmdGcN8FH--ys3ELbD1/view> Topics covered: 1. Transition component of HeadlessUI and its advantage 2. Dialog Component of HeadlessUI and its advantage. 3. Used Transition and Dialog component to create Starbucks Rewards page sidebar menu. 4. Directions on how to create a confirmation dialog using HeadlessUI.

First 17 minutes or so of the video are blank. So please skip those while watching/rewatching

Lecture 21 (7th July, 2021) -

<https://drive.google.com/file/d/19rZYdHkuPKDDYstNU9YO16iLDa-dvhZY/view> Topics covered: 1. Why key is need while looping over elements i.e. React state mismatch issue. 2. Starbucks rewards page like Tabs using Tailwind CSS. 3. Convert Tailwind code into Reusable React Components.

Lecture 22 (8th July, 2021) -

<https://drive.google.com/file/d/1NjxoPmBFMGsUcWMZvn7-iF33L-sLGGST/view> Topics covered: Explanation of how we built Reusable Tab component and why we took that route.

Lecture 23 (9th July, 2021) -

https://drive.google.com/file/d/1gU32pYQKuEXzJdqTqzVETZF_fgUVYxYU/view Topics covered: 1. Default value in generics. 2. Use of translate over left etc for animation to improve efficiency. 3. React.XYZ v.s. direct importing and using XYZ as a named import. 4. rel="noopener" on <a> tags and it's advantage in security. 5. Event loop and async

tasks. 6. Very basics of using Axios and fetch to get data from server. 7. useEffect and it's use.

Lecture 24 (12th July, 2021) -

<https://drive.google.com/file/d/1nPSLcXwm2Oi8-i-CQzro29Clgf7yQlVL/view> Topics covered: 1. useEffect and useLayoutEffect sync or async? (Please note that we will cover useEffect and useLayoutEffect further in future lectures) 2. Map, Filter, Reduce on Arrays 3. Promises vs callbacks 4. Promises - then and catch chaining.

Lecture 25 (13th July, 2021) -

<https://drive.google.com/file/d/1bzCLlrAEhGLioCYV4qlAEzbaI3GQZIXN/view> Topics covered: 1. Promises continued... 2. Async Await.

Lecture 26 (14th July, 2021) -

https://drive.google.com/file/d/18VDAjuW6rGUZK74y_MEmzIsuFGBJEMbQ/view Topics covered: 1. How project lifecycle works. 2. Started React Project Development. 3. Why nested routing is needed and how to do it. 4. Why routes with Params are needed and how to use them.

Lecture 27 (15th July, 2021) -

https://drive.google.com/file/d/1YI1lck14ts3gcGcJlDmmxg-p3_s9OzrC/view Topics covered: 1. autoComplete attribute for inputs to support auto fill by browsers/extensions. 2. Concept of request rewrite to index.html when file is not found. 3. Form handling in React. 4. Input validation. 5. Managing touched state of Inputs. 6. Managing submitting state. 7. On successful login, redirection using React router's useHistory hook. 8. Using Yup of validation.

Lecture 28 (16th July, 2021) -

<https://drive.google.com/file/d/1nhnm4P63RpsG6l5LqAaOsPdAHKj8aYWD/view> Topics covered: 1. Yup for form validation continued. 2. Discussion on Hooks, Custom Hooks and useEffect. 3. Using formik for form validation.

Lecture 29 (20th July, 2021) -

<https://drive.google.com/file/d/1nCPFvuKi9kPGtQ6b9i9A1npzOBJ-UrXS/view> Topics covered: Storybook

Lecture 30 (21th July, 2021) -

https://drive.google.com/file/d/1gMHN4oNZml4E286r_VIEvtcdQlvV44jp/view Topics covered: 1. Deployment of React app and storybook on Netlify. 2. REST APIs

Lecture 31 (22nd July, 2021) -

<https://drive.google.com/file/d/1Y5wantsCtBRsn6fy8bjTTXYYAJyRQjQ-/view> Topics covered: 1. Image src - imports vs direct urls. 2. Need of string formats for Data transfer during API calls. 3. Overview of JSON format. 4. Sending data in body vs url vs headers.

Lecture 32 (23rd July, 2021) -

<https://drive.google.com/file/d/1Z4FQK-BWcL5cPQVj9bPbJLekRus6vwij/view> Topics covered: 1. HTTP - stateless protocol and cookies. 2. Options request and CORS. 3. Why we don't necessarily need Cookies when sending requests using JS. 4. LocalStorage 5. Conditionally redirecting to different pages based user login status. 6. Using axios request interceptor to send auth token in every request. 7. Using axios response interceptor to clear token and redirect to login page in case "Invalid Token"

Lecture 33 (26th July, 2021) -

<https://drive.google.com/file/d/15F567kbhCueX7OhldbKXplEzmVX8Clkl/view> Topics covered: 1. Storing auth token in a JS variable vs localStorage vs sessionStorage. 2. Short Discussion on security and things like Cross-Site Request Forgery (CSRF), Cross-origin Resource Sharing (CORS) etc. 3. Refactored api.ts code to make it more manageable as project grows

Lecture 34 (27th July, 2021) -

<https://drive.google.com/file/d/1Hs0PPyF1WsneBosymhF0xJdrWqDpVrwF/view> Topics covered: 1. Lazy loading with Suspense. 2. Storing common data like logged in user object globally. 3. Updating and using global state by passing value and update function in props. (This is a naive way and we will improve on it in next class)

Lecture 35 (28th July, 2021) -

<https://drive.google.com/file/d/1QzEoyYOQ09jIN8I8ueFJsCq7FpS4lpkO/view> Topics covered: 1. Bit of discussion on Hooks. 2. Why we need something like React Context. 3. Using React Context.

Lecture 36 (29th July, 2021) -

<https://drive.google.com/file/d/11z59sW18aDfppjKnpDJ1hbju2RluPyZU/view> Topics covered: 1. useCallback, useMemo 2. generator functions.

Lecture 37 (30th July, 2021) -

<https://drive.google.com/file/d/1yMM5SSxJkgtTFqORrxsb-1qG2zbhSeS3/view> Topics covered: 1. Why we need Redux 2. Basics of Redux

Lecture 38 (2nd Aug, 2021) -

<https://drive.google.com/file/d/1dBFyi5svc4wi-U-oEJOUY0ApseHEfTel/view> Topics covered: 1. Why mutation is important in Redux? 2. Why can't we start with a clone of current state in reducer and then mutate it? 3. Storing data in normalised form in Redux. Why and how?

Lecture 39 (3rd Aug, 2021) -

https://drive.google.com/file/d/1Lc7yOsoX9cP2XYt7TLRr-lvBve_OoL4Z/view Topics covered: 1. Closures in JavaScript 2. Accessing object properties with dot vs []

Lecture 40 (4th Aug, 2021) -

https://drive.google.com/file/d/1ftmkjT_JGUQ_F7PuWRrEt8KLFAklfdEG/view Topics covered: 1. Sub reducers in Redux. 2. using bindActionCreators.

Lecture 41 (5th Aug, 2021) -

https://drive.google.com/file/d/1ZCnXQ--VKYOVUV0_5QTpkEG69KdG9HWD/view Topics covered: 1. ESLint 2. Bit more about hooks and custom hooks. 3. Creation of memoized selectors using createSelector.

Lecture 42 (6th Aug, 2021) -

<https://drive.google.com/file/d/129WV2N5RdQrrtAgMQMGO-DjKxNJl26Kx/view> Topics covered: 1. Bit more on createSelector and it's important. 2. Refactored reducer code to use EntityState and Entity model.

Lecture 43 (9th Aug, 2021) -

<https://drive.google.com/file/d/1VqVqAEwYThzU5CbSm2iG56QsQYntWcon/view> Topics covered: 1. How to decide when to store data in Redux vs useState. 2. Storing loading state in redux with and without previous request cancelation approach. 3. How to cancel request with axios.

Lecture 44 (10th Aug, 2021) -

<https://drive.google.com/file/d/1Uyr5ITZlr83QjwWt2W6SYLMWRR9M87cl/view> Topics covered: 1. Redux Saga 2. Bit more about generator function.

Lecture 45 (11th Aug, 2021) -

<https://drive.google.com/file/d/17AXLpgJWLUJ06yju0BGznRknqmVNO3MD/view> Topics covered: 1. Revision of why Redux works the way it works. (Kaahe itna ghuma phira kar

kaan pakadte hain) 2. Why Redux saga works the way it works. 3. How to do debounce and cancel old requests with Saga and axios in a reusable way.

Lecture 46 (12th Aug, 2021) -

<https://drive.google.com/file/d/1yFbqOfyUhugnrAXHmB1e4AmKMW7ByRa0/view>

Topics covered: 1. Doubts related to Redux and Saga 2. Implementation of Group detail page.

Lecture 47 (13th Aug, 2021) -

<https://drive.google.com/file/d/1Deo5KhJ0pGLo7uAbtXNBJ8WjOlXqMtEz/view> Topics

covered: 1. Summary of steps to follow to implement a feature with Redux and Saga. 2. Loading state handling with Saga on Group Detail page. 3. Error handling with Saga.

Lecture 48 (16th Aug, 2021) -

https://drive.google.com/file/d/14tq3QYB74rNzIC_WG0cQDQx3wTN07Gaf/view Topics

covered: 1. Storing one-to-one, one-to-many and many-to-many relationships in Redux. 2. Type Conversion v.s. Type Casting. 3. Quick overview of Lodash utility library.

Lecture 49 (17th Aug, 2021) -

https://drive.google.com/file/d/1YTZvvLWfOXamHLbo5V_gRx1Vz1LKqAAN/view Topics

covered: Implementation of Redux relationship assignment.

Lecture 50 (18th Aug, 2021) -

<https://drive.google.com/file/d/1GZQbl7sJwLDVpsjJhAkzdRBPk-UfC3LY/view> Topics

covered: 1. Student doubts related to Redux and Saga. 2. Difference between blocking and non blocking Saga Effects. 3. Introduction to few Saga effects like fork, take, spawn.

Lecture 51 (19th Aug, 2021) -

<https://drive.google.com/file/d/1VSNKXTZaTkU0f1E1qIMbm4Z0HD4fUwgR/view> Topics

covered: 1. Wrapping functions that accept callbacks to create functions that return promise. 2. `yield` in Generator functions v.s. `await` in async function. 3. Some more Saga effects.

Lecture 52 (23rd Aug, 2021) -

https://drive.google.com/file/d/1uho1ZTvoLgBKArxS_1EkaQu9FFkkXJnn/view Topics

covered: 1. Async await and loops. 2. Create a generic promisify function. 3. Higher Order Components (HOCs). 4. Further discussion on Normalizr and how to handle entity relationships in Redux

Lecture 53 (24th Aug, 2021) -

https://drive.google.com/file/d/1_fSfGdgAzu0CH6BpK1c98aldBGmIU_R3/view Topics covered: 1. How new Promise() works behind the scene. 2. Using connected-react-router and maintaining route state in Redux. 3. Using Location_updated action to move logic from component to saga. 4. Significance of generics passed to Generator function

Lecture 54 (25th Aug, 2021) -

https://drive.google.com/file/d/1_nbMMzAkd2h4s8VO_ho3MpVOzOQL3lAV/view Topics covered: 1. Moving from hardcoded groups/23 to groups/:id in locationChangeSaga. 2. Using createMatchSelector to create selectors for selectedEntityId where Entity could be any entity like Group. 3. Git flow. 4. Rendering Props 5. Different approaches that Formik provides i.e. hooks, context+hooks, rendering props and HOC.

https://drive.google.com/file/d/1i3UZotPi_IB6U9jXZECj6CaDltgM0ODA/view Topics

covered: 1. Class components and lifecycle methods 2. PureComponent 3. Redux-react connect() HOC. 4. Ref, createRef and useRef.

Lecture 56 (31st Aug, 2021) -

https://drive.google.com/file/d/1fR68N6JxLgPPCCwh_UGnOv6upsQuG8nv/view Topics covered: 1. Clarification on useLayoutEffect vs useEffect 2. Ref forwarding using props like inputRef or myDivRef 3. Ref forwarding using ref and forwardRef HOC.

Lecture 57 (2nd Sep, 2021) -

<https://drive.google.com/file/d/1Zztbg7kHSv5Ma0zN9lyQTm7iPLO1i0Ap/view> Topics covered: 1. Error Boundaries 2. React portals